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Introducing Figure 03

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Today we're introducing Figure 03, our 3rd generation humanoid robot. Figure 03 is designed for Helix, the home, and the world at scale. Our goal is to deliver a truly general-purpose robot - one that can perform human-like tasks and learn directly from people. To realize this vision, our engineering and design teams completed a ground-up hardware and software redesign to ship Figure 03 for:

Helix: Figure 03 features a completely redesigned sensory suite and hand system which is purpose-built to enable Helix - Figure's proprietary vision-language-action AI.

The home: Figure 03 has several new features, including soft goods, wireless charging, improved audio system for voice reasoning, and battery safety advancements that make it safer and easier to use in a home environment.

Mass manufacturing: Figure 03 was engineered from the ground-up for high-volume manufacturing. In order to scale, we established a new supply chain and entirely new process for manufacturing humanoid robots at BotQ.

The world at scale: The lower manufacturing cost and the advancements made for Helix have significant benefits for commercial applications.

Designed for Helix

There's no path to scaling humanoid robots without AI. That's why we built Figure 03 around one goal - to enable true reasoning throughout the world using Helix. Figure 03 introduces a fully redesigned sensory suite and hand system, purpose-built to bring Helix to life.

Figure 03 introduces a next-generation vision system engineered for high-frequency visuomotor control. Its new camera architecture delivers twice the frame rate, one-quarter the latency, and a 60% wider field of view per camera - all within a more compact form factor. Combined with an expanded depth of field, this architecture provides Helix with a denser, more stable perceptual stream. These advancements are essential for intelligent navigation and precise manipulation in complex, cluttered spaces such as homes.

Each hand now integrates an embedded palm camera with a wide field of view and low-latency sensing, which offers redundant, close-range visual feedback during grasps. These cameras allow Helix to maintain visual awareness even when the main cameras are occluded (i.e. when reaching into a cabinet or working in confined spaces) and enable continuous, adaptive control in real time.

The Figure 03 hands represent a major leap in compliant and tactile design. Softer, more adaptive fingertips increase surface contact area, enabling more stable grasps across objects of varied shapes and sizes. After surveying existing market options, Figure found that current tactile sensors had inherent limitations that could not withstand real-world use. This led to the internal development of our first-generation tactile sensor, guided by three principles: extreme durability, long-term reliability, and high-fidelity sensing.

Each fingertip sensor can detect forces as small as three grams of pressure - sensitive enough to register the weight of a paperclip resting on your finger. This precision enables Helix to distinguish between a secure grip and an impending slip before it occurs, allowing fine-grained, dexterous control over fragile, irregular, or moving objects.

Figure 03 also includes 10 Gbps mmWave data offload capability, allowing the entire fleet to upload terabytes of data for continuous learning and improvement. Together, these advancements position Figure 03 as uniquely capable of large-scale, end-to-end pixels-to-action learning.

Designed for the Home

To operate effectively in the home, a robot must work seamlessly alongside people in their daily environments. With this in mind, Figure 03 introduces several design improvements focused on safety. It features strategically placed multi-density foam to protect against pinch points, and is covered in soft textiles rather than hard machined parts. Figure 03 also has 9% less mass and significantly less volume than Figure 02, making it easier to maneuver through household spaces.

The Figure 03 battery pushes the bounds for robot battery safety and incorporates multiple layers of protection against abuse or malfunction, including safeguards at the Battery Management System (BMS), cell, interconnect, and pack levels. The battery has already achieved certification to the UN38.3 standard.

Beyond safety, Figure 03 is designed for everyday usability. The soft goods are fully washable and can be removed or replaced without tools, allowing quick and easy swaps. The robot can also be customized with various clothing options, including garments made from cut-resistant and durable materials.

To make it easier to communicate naturally with the robot, Figure 03 features an upgraded audio hardware system for better real time speech-to-speech. Compared with Figure 02, its speaker is twice the size and nearly four times more powerful, while the microphone has been repositioned for improved performance and clarity.

Continuing our vision for a fully autonomous, wire-free system, Figure 03 is capable of wireless inductive charging alongside wireless data offload. Charging coils in the robot's feet allow it to simply step onto a wireless stand and charge at 2 kW. In a home setting, this means the robot can automatically dock and recharge itself as needed throughout the day.

Designed for Mass Manufacturing

Humanoid robots have traditionally been designed as engineering prototypes which are time consuming and expensive to produce. Figure 03 is our first robot engineered from the ground-up for high-volume manufacturing. We achieved this through three major initiatives:

Design and process reinvention

Establishing an entirely new supply chain

The invention of BotQ, our high-volume manufacturing facility

Moving from Figure 02 to Figure 03 required redesigning nearly every component of the robot with manufacturability and cost in mind. The mechanical and electrical engineering teams aggressively reduced part count, assembly steps, and any components that were not absolutely critical to meet design requirements. While Figure 02 was primarily designed to be manufactured with CNC machining, Figure 03 relies heavily on tooled processes such as die-casting, injection molding, and stamping. This shift demanded a significant up-front investment in tooling, but the payoff is clear: each Figure 03 unit now costs dramatically less to build, with the economics improving as volumes grow.

To scale Figure 03, Figure had to build an entirely new supply chain for an industry where one does not currently exist. Figure chose to vertically integrate across many critical module builds including actuators, batteries, sensors, structures, and electronics, all of which were designed completely in-house. For individual components, Figure strategically identified and partnered with suppliers capable of meeting the required volumes, timelines, and strict quality standards demanded by the team. The

result of this year-long effort is a global network of partners who can grow alongside Figure and meetproduction goals of thousands and eventually millions of parts under an aggressive ramp schedule.

BotQ is Figure's dedicated manufacturing facility designed to scale robot production. BotQ's first-generation manufacturing line will initially be capable of producing up to 12,000 humanoid robots per year, with the goal of producing a total of 100,000 robots over the next four years. Instead of relying on contract manufacturers, Figure brought production of its most critical systems in-house to maintain tight control over quality, iteration, and speed. The facility is equipped with state-of-the-art systems and digital integrations, anchored by our internally developed Manufacturing Execution System (MES). Every subassembly and final assembly passes through this line with full traceability, ensuring quality, repeatability, and continuous improvement.

Designed for the World at Scale

Figure's focus on the home market in no way detracts from the potential of Figure 03 for the commercial market. By solving for the variability and intractability of the home, Figure is developing a truly general-purpose product that can do the widest possible range of tasks in the workforce.

Figure 03 is well suited for commercial applications for several reasons. The actuators can perform at 2x faster speeds with improved torque density (nm/kg). The most significant result of this is our ability to pick and place items at faster speeds.

The improvements to the hands and sensory suite made for Helix are of major significance for commercial use cases. With the camera and perception system upgrades, Figure 03 will be able to intelligently navigate commercial environments and execute precise manipulation. The changes to the hands highlighted above (added compliance, fingertip surface area, tactile sensing) enable better and more stable grasps across an array of objects such as small pieces of sheet metal and deformable poly bags.

Thanks to inductive charging, Figure 03 is capable of near-continuous operation as long as it can step onto a charging mat for a certain period of time during the use case. The fast wireless data offload also means that the robot can offload seamlessly during shift breaks just by returning to the dock.

Commercial customers can also design distinct uniforms for their Figure 03 fleet, with the option to use more durable, or cut-proof materials, and make other design changes for specific environments. New side screens on Figure 03 even allow quick identification across large fleets and can be fully customized to match each customer's branding or operational needs.

Conclusion

Figure 03 represents an unprecedented advancement in taking humanoid robots from experimental prototypes to deployable, scalable products. By uniting advanced perception and tactile intelligence with home-safe design and mass-manufacturing readiness, Figure has built a platform capable of learning, adapting, and working across both domestic and commercial settings. Designed for Helix, the home, and the world at scale, Figure 03 establishes the foundation for true general-purpose robotics - one capable of transforming how people live and work.

What's on your mind?

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